

Cryptography and System Security Lab – Answers

Experiment 1

1. A substitution cipher is an encryption technique where each letter of the plaintext is replaced with another letter according to a fixed rule.
2. Plaintext alphabet is the original set of letters (A–Z) used before encryption.
3. Ciphertext alphabet is the substituted alphabet used to replace plaintext letters during encryption.
4. Example: Caesar Cipher where A becomes D, B becomes E, etc.
5. Advantages: Easy to implement, fast encryption, simple concept.
6. Transposition cipher is a method of encryption where the positions of characters in plaintext are rearranged according to a system.
7. Purpose: To increase security by changing the order of characters without altering the characters themselves.

Experiment 2

1. RSA is a public key cryptography algorithm used for secure data transmission using two keys: public key and private key.
2. Asymmetric cryptography means encryption uses two different keys: one for encryption and one for decryption.
3. Two branches: Public key encryption and Digital signatures.
4. Digital signatures are used to verify the authenticity and integrity of a message.
5. Encryption: $C = P^e \bmod n$ Decryption: $P = C^d \bmod n$

Experiment 3

1. Diffie–Hellman is a method used to securely exchange cryptographic keys over a public channel.
2. Symmetric key: Same key used for both encryption and decryption.
3. Asymmetric key: Uses two keys – public key and private key.
4. Need: To securely generate a shared secret key between two parties over an insecure network.
5. Working: Alice and Bob choose a prime number and base, generate public values using private numbers, exchange them, and compute the same shared secret key.

Experiment 4

1. MD5 is a cryptographic hash function that produces a 128-bit hash value used for verifying data integrity.
2. SHA (Secure Hash Algorithm) is a family of cryptographic hash functions used to generate secure hash values for data integrity and security.

Experiment 5

1. Network reconnaissance tools: WHOIS, Dig, Traceroute, Nslookup.

2. Traceroute is a command used to track the path packets take from source to destination. TTL (Time To Live) limits the number of hops a packet can travel.
3. Dig command is used to query DNS servers and retrieve domain information such as IP addresses.
4. Nslookup is used to find the IP address associated with a domain name.
5. IP Address: Unique identifier for devices on a network. Static IP is fixed; Dynamic IP changes automatically.
6. Whois command is used to retrieve information about domain ownership and registration details.
7. Network configuration files: /etc/hosts, /etc/resolv.conf, /etc/network/interfaces, /etc/hostname.

Experiment 6

1. Packet sniffer is a tool used to capture and analyze network packets.
2. Install Wireshark: `sudo apt-get install wireshark`.
3. Use: To monitor network traffic and diagnose network problems.
4. Wireshark is a network protocol analyzer used to capture and analyze packets.
5. Install tcpdump: `sudo apt-get install tcpdump`.
6. Use of tcpdump: Command-line packet analyzer used to capture network traffic.
7. Ethereal is the old name of Wireshark.

Experiment 7

1. NMAP is a network scanning tool used to discover hosts and services on a network.
2. Install NMAP: `sudo apt-get install nmap`.
3. Port scanning is the process of checking open ports on a host.
4. TCP port scan identifies open TCP ports on a system.
5. UDP port scan identifies open UDP ports on a system.
6. TCP scan is more reliable while UDP scan is slower and less reliable.
7. NMAP works by sending packets to target hosts and analyzing responses to determine open ports and services.

Experiment 8

1. IPSEC is a protocol suite used to secure Internet communications through encryption and authentication.
2. IPSEC configuration involves installing IPSEC tools and configuring policies and keys.
3. IPSEC protocols: AH (Authentication Header) and ESP (Encapsulating Security Payload).
4. Authentication Header provides integrity and authentication of IP packets.
5. Encapsulated Security Payload provides encryption, authentication, and integrity.
6. IKE (Internet Key Exchange) protocol is used to establish secure connections and exchange keys.
7. IPSEC is used to create secure VPN connections over the internet.

Experiment 9

1. Firewalls are network security systems that monitor and control incoming and outgoing network traffic.
2. Built-in chains: INPUT, OUTPUT, FORWARD.
3. Example commands:

4. iptables -L : List firewall rules.
5. iptables -A : Append rule to a chain.

Experiment 10

1. SQL Injection is a web attack where malicious SQL queries are inserted into input fields to manipulate databases.
2. Cross Site Scripting (XSS) is an attack where malicious scripts are injected into web pages and executed in the user's browser.